

BANK LOAN REPORT ANALYSIS

Data Analytics Project Documentation

1. Introduction

The Bank Loan Report Analysis project focuses on evaluating the performance of a bank's lending operations using data analytics and visualization techniques. The project analyzes loan applications, funded amounts, repayments, borrower profiles, and risk indicators to generate meaningful business insights.

Using PostgreSQL, Power BI, and Python, the project helps in tracking loan portfolio health, identifying trends, and distinguishing between good and bad loans. The final output is an interactive dashboard that supports data-driven decision-making in banking operations.

2. Problem Statement

The objective of this project is to analyze and monitor the bank's lending operations using data analytics tools. The goal is to evaluate loan applications, funded amounts, repayment behavior, borrower profiles, and financial risk indicators.

This analysis helps:

- Track loan portfolio performance
 - Identify lending trends
 - Evaluate borrower risk
 - Distinguish between good and bad loans
 - Support data-driven business decisions
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3. Dataset Information

The dataset contains bank loan records including:

- Loan application details
 - Borrower demographic information
 - Loan status
 - Funded amount and received amount
 - Interest rate and installment details
 - Employment and income details
 - Credit-related attributes
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4. Technologies Used

- PostgreSQL (Data extraction and SQL analysis)
 - Power BI (Dashboard creation and visualization)
 - Python (Data analysis and EDA)
 - Pandas (Data manipulation)
 - Matplotlib (Data visualization)
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5. Key Performance Indicators (KPIs)

The following KPIs are used to analyze loan performance:

- Total Loan Applications
- Total Funded Amount
- Total Amount Received
- Average Interest Rate
- Average Debt-to-Income Ratio (DTI)

Additional tracking:

- Month-to-Date (MTD) values

- Month-over-Month (MoM) changes
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6. Dashboard 1: Summary Dashboard

Objective

The Summary Dashboard provides a high-level overview of the bank's lending performance and key financial metrics.

Key Analysis

Good Loan vs Bad Loan Analysis

Loans are classified into:

Good Loans

- Fully Paid
- Current

Bad Loans

- Charged Off
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Good Loan KPIs

- Good Loan Percentage
- Total Good Loan Applications
- Good Loan Funded Amount
- Good Loan Amount Received

Bad Loan KPIs

- Bad Loan Percentage
 - Total Bad Loan Applications
 - Bad Loan Funded Amount
 - Bad Loan Amount Received
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Loan Status Grid View

This section provides a complete summary of loan performance including:

- Total Loan Applications
 - Total Funded Amount
 - Total Amount Received
 - MTD Funded Amount
 - MTD Amount Received
 - Average Interest Rate
 - Average DTI
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7. Dashboard 2: Overview Dashboard

Objective

The Overview Dashboard visualizes lending trends and borrower behavior using charts and graphs.

Visualizations

1. Monthly Trends (Line Chart)

- Loan Applications

- Funded Amount
 - Amount Received
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2. Regional Analysis (Map Chart)

- State-wise loan distribution
 - Funded and received amounts by region
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3. Loan Term Analysis (Donut Chart)

- 36 months
 - 60 months
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4. Employee Length Analysis (Bar Chart)

- 1 year
 - 5 years
 - 10+ years
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5. Loan Purpose Breakdown (Bar Chart)

- Debt consolidation
 - Credit card refinancing
 - Home improvement
 - Business loans
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6. Home Ownership Analysis (Tree Map)

- Own

- Rent
 - Mortgage
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8. Dashboard 3: Details Dashboard

Objective

The Details Dashboard provides a complete view of all loan records in a structured format.

It helps users:

- View borrower-level data
- Analyze loan status
- Track repayment behavior
- Access detailed financial information

This dashboard acts as a central data explorer for the entire loan dataset.

9. Python Analysis (Brief)

Python was used for:

- Data cleaning
 - Exploratory Data Analysis (EDA)
 - Visualization of trends
 - Understanding distributions of key variables
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10. SQL Analysis (Brief)

SQL was used for:

- KPI calculations

- Aggregation queries
 - Filtering loan status data
 - Generating summary tables for Power BI
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11. Key Insights

- Majority of loans are issued for debt consolidation purposes
 - Good loans form a major portion of the portfolio
 - Borrowers with higher DTI show higher risk of default
 - Certain states show higher lending activity
 - Loan purpose strongly affects repayment behavior
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12. Conclusion

The Bank Loan Report Analysis project successfully demonstrates how data analytics can be applied in the banking sector to evaluate loan performance and manage financial risk.

By using PostgreSQL, Power BI, and Python, the project provides meaningful insights into borrower behavior, loan quality, and portfolio performance. These insights help banks make better lending decisions and improve financial stability.